The Price of a Flat Direction: Transverse Curvature Sets Retention in a Trained Recurrent Memory

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Abstract

A manifold of marginally stable states stores a continuous value, and it is known to be fragile: noise diffuses the state along it, and perturbing the dynamical law — learning included — destroys it. What has not been available is the exchange rate. We supply one. Our object is a recurrent unit advancing a latent state (q, p) by a damped symplectic velocity-Verlet step of a learned Hamiltonian. Its trained potential V_θ is G-invariant, its minimisers form a G-orbit, the curvature transverse to that orbit sets how long a written value survives, and friction and temperature move the state along it. Retention half-life scales as $n_{1/2} \propto \mu^{-2}$ in the transverse curvature μ — verified on trained models at latent dimension 4 on an S^1 testbed, ≤ 5 seeds, laptop CPU, exact to $1.000000 \pm 5\times10^{-12}$ over 4.5 decades — and it saturates at a curvature-independent floor past a crossover at $\varepsilon\mu \approx \gamma/2$. Three consequences follow. (i) A recently posted machine-learning lifetime law is this budget's overdamped face: run unchanged on our trained units it matches to $\leq 3\%$, and mispredicts past the crossover in a calculable direction. (ii) A linear spurion on 8 trained $SO(2)$ checkpoints makes the three objects of $\mu^2 F^2 = \delta\Sigma$ separately measurable. (iii) Designed protection beats emergent protection by 13–14 orders of magnitude in μ^2 on this architecture class, and every headline law survives the training-time correction that keeps the orbit intact. Designed-testbed results are labeled verification; learned-system results, evidence.

1 Introduction

A recurrent state holding an analog value can be built on a flat direction: a set of states the dynamics do not distinguish, so a written value neither decays nor is pulled to a preferred point. Symmetry supplies such directions for free. If a learned potential V_θ is G-invariant and its minimiser is not, the minimisers form a G-orbit, the Hessian annihilates every orbit tangent, and the orbit coordinate is neutral. *However*, a flat direction is only as useful as its price list, and none existed for a trained model: how much retention a given transverse curvature buys, where the law stops holding, and whether it survives the correction that keeps the orbit alive. *We measure all three.* The object advances (q, p) by a symplectic velocity-Verlet step of a learnable separable Hamiltonian $H(q, p) = T(p) + V_\theta(q)$, with a per-step damping $p \mapsto (1 - \gamma)p$ supplying controllable forgetting. This is not a new architecture but a closed-form account of retention on a trained one. The reference unit is the CLU (Causal Learning Unit), introduced as CHLU in Jawahar & Pierini (2026); results hold for the class of damped symplectic recurrences, and the exactly-solvable theory verified here is developed in a separate note, cited as *the theory note* (Anonymous, 2026, in the supplementary material).

Two masses, and one conversion. Two quantities here are called "mass" and run in opposite directions. The inertial mass M_i is the learned diagonal of T: larger M, slower. The spectral mass

μ_k, which we lead with as the *transverse curvature*, is $\mu_k^2 = \lambda_k(M^{-1/2}\nabla^2 V_\theta M^{-1/2})$ at a critical point: larger μ, shorter memory. All retention statements use μ. One conversion precedes any cross-field comparison: flow-Jacobian eigenvalues are inverse time, whereas μ^2 are eigenvalues of a mass-whitened Hessian of a *potential*, inverse time squared. Lifetimes are half-lives $n_{1/2}$ in map applications, not τ and not diffusion coefficients D; each is reported with the metric defining it, because those metrics bifurcate past the crossover.

Contributions. (1) The transverse-curvature budget, verified on trained models (Appendix D): the μ^{-2} law at overdamped slope -0.985 against a predicted -1, the curvature identity to 5×10^{-12} over 4.5 decades, the curvature-independent floor and the crossover's $\sqrt{h-h^*}$ frequency onset on 5 designed seeds at dim 4, and the condensate-resolving spurion on 8 trained $SO(2)$ checkpoints. (2) The head-to-head against Mo's lifetime law on trained checkpoints (§4.1), reproduced on Mo's own estimator. (3) The symmetry-realization taxonomy exhibited by trained models (§3), the designed-versus-emergent gap and the price of the physics (§4.2). (4) The absence of the retention triad in learned baselines, and the recipe that keeps a designed orbit intact (§4.3). Designed-testbed results — invariant potentials, analytic tilts and spurions — are verification of the theory's exactness; learned-system, training-dynamics and head-to-head results are evidence.

2 Related work

Neutrality of group-orbit directions is classical (Golubitsky, Stewart & Schaeffer 1988; Krupa 1990; Rumberger 2001). The same object is studied as a *continuous attractor*: a manifold whose points are all fixed points, "marginally stable tangent to the manifold and stable normal to the manifold" (Ságodi et al. 2024), the substrate of the neural integrator (Seung 1996) and of measured ring and toroidal codes (Kim et al. 2017; Gardner et al. 2022; Khona & Fiete 2022). We make no claim about biological systems and do not model neural data. Four facts there are established, and none is ours. Noise diffuses the coordinate along the manifold, the diffusion coefficient lower-bounded by the code's own Fisher information (Burak & Fiete 2012). The manifold is structurally unstable, destroyed by most infinitesimal changes of the dynamical law, with online learning named as one perturbation source — the *fine-tuning problem*, whose memory error Ságodi et al. bound by the unknown perturbation's size, worst-case and linear in t. A homeostatic mechanism restoring a destroyed flat direction is 23 years old (Renart, Song & Wang 2003), so a corrective term is not our idea. And a local learning rule can *produce* accurate ring-attractor tuning (Vafidis et al. 2022), so nothing we measure implies that learning cannot build a flat direction. Absent is the exchange rate: how much retention a given transverse curvature buys, where that law stops holding, and whether it survives the correction. We measure all three on trained models, in closed form, with a regime map — floor, crossover — that a first-order flow cannot exhibit.

Closest to us, Mo (2026) proves that a C^1 field exactly equivariant under a Lie group has at least $\dim(G/\mathcal{H})$ zero Lyapunov exponents tangent to the orbit; we use his diagnostics and his breaking-and-censoring protocol. His result is the *kinematics* of protection, silent on what sets the gap; here the gap is *constitutive*, and Lyapunov spectra are blind to the write semantics — transport, latch, noise accumulation — that make a neutral direction a *memory*. His theorem states sufficiency; equivariance is not necessary, the latch being a modulus of V_θ. Constructive equivariant memory (Di Bernardo et al. 2025; Keller 2025; Iqbal, Keller, Song, Miyato & Welling 2026) engineers *which* symmetry a network expresses but does not price *time*: we claim no novelty for choosing a symmetry group or coset, only the price list attached to a chosen structure. Four further neighbours are this audience's own: exact versus relaxed equivariance as a design axis (van der Ouderaa & van der Wilk 2023); reading a landscape by its level sets and critical points (Akhtiamov & Thomson 2023; Wang & Ponce 2023, on tuning landscapes over stimuli, not latent states); a Lie-group representation carried in a recurrent state on a torus (Xu et al. 2023); and a quotient coordinate as an input encoding (Aslan, Platt & Sheard 2023), where ours is a storage device. Slow structure *appearing* during training has a recent analytical model (Dinc et al. 2025, preprint); §4.3 runs the opposite direction. Cross-session representational drift (Driscoll et al. 2017; Aitken et al. 2022; Jude et al. 2023) is a different phenomenon from motion along a flat direction and we do not address it; the only motion we report is of the coset coordinate under our own dynamics. Our comparators are LSTM (Hochreiter & Schmidhuber 1997), LEM (Rusch et al. 2022) and coRNN (Rusch & Mishra 2021a); conformal symplecticity is McLachlan & Perlmutter (2001), the leapfrog $h < 2$ limit standard (Hairer, Lubich

& Wanner 2006), and wake–sleep (Hinton et al. 1995) with PCD (Tieleman 2008) the contrastive objective, whose landscape distortion is classical (Fischer & Igel 2010; Nijkamp et al. 2020).

Four claims we retire, so that a reviewer's first objections are answered in writing (elaborated, with their numbers, in Appendix C). *(1)* Attention and state-space models *do* have retention-guarantee analogues, and HiPPO-LegS (Gu et al. 2020) is the counterexample. *(2)* "Retrieval is a rollout, not a weighted sum" is Kong, Brewer & Lai's (2024) move, not our novelty. *(3)* Continuity is no escape hatch from the NTM/DNC lineage (Graves et al. 2014, 2016; Csordás & Schmidhuber 2019), which was already fully soft. *(4)* This is no competitor to the transformer, and was never claimed to be. Our claim is correspondingly narrow: a physics-structured associative memory in which retention is a designed, per-item property — permanent (symmetry-protected, $\mu^2 \equiv 0$ exactly) or of a finite half-life set by μ^2 — rather than an emergent consequence of a learned recurrence — *controllability*, which we evidence, not *capability*, which we do not claim against transformers or state-space models. The nearest published neighbour to our write is Titans (Behrouz et al. 2025), test-time gradient descent with momentum on an associative-memory loss: the same family of discretized damped second-order dynamics, differing in that our damping and inertia are *read out and priced*, not optimizer hyperparameters.

3 Setup, and the two axes

The map, and the three bands. One dissipative velocity-Verlet step with step ε and damping γ advances (q, p) by a kick–drift–kick of $H = T(p) + V_\theta(q)$ followed by $p \mapsto (1-\gamma)p$. It is conformally symplectic — $J^\top \Omega J = (1-\gamma)\Omega$, $\det J = (1-\gamma)^d$: phase-space volume contracts by a fixed factor per step, so the geometry is preserved and only the scale is spent. Memory at each normal mode reduces to one 2×2 block ($\mu^2 = k/m$, $h = \varepsilon\mu$) with three bands: a latch ($\mu = 0$, symmetry-protected) with frozen displacement $q_\infty = q_0 + \varepsilon p_0/(M\gamma)$ and infinite half-life; an overdamped register ($0 < \varepsilon\mu \lesssim \gamma/2$) at $n_{1/2} \approx 2\gamma \ln 2/[(2-\gamma)(\varepsilon\mu)^2]$; and an underdamped working memory ($\gamma/2 \lesssim \varepsilon\mu < 2$) at the curvature-independent envelope half-life $2\ln 2/(-\ln(1-\gamma))$. The boundary is the exceptional point $h^* \approx \gamma/2$, the overdamped-to-underdamped crossover at which the block becomes defective. Relaxation onto the orbit is the fast normal flow; retention is the slow tangent flow. One difference from a classical neural integrator we do not smooth over: an integrator stores by *not* damping, whereas at $\gamma = 0$ our flat mode never freezes a write and any $\gamma > 0$ gives an exact latch. Trained models are the reference unit's Experiment-D configuration — dim 4, hidden 64, learned-Newtonian kinetic term, $\mathrm{dt} = 0.05$, unit circle at 256 points, 150 epochs (§4.3 for why not the default 1000) — in two families: designed, an architecturally-$SO(2)$-invariant V_θ on 5 seeds, and emergent, a generic MLP V_θ on the same data on 3 seeds.

Axis 1: which symmetry the trained potential realizes. It fixes each mode's μ^2, a statement about V_θ alone, in three cells, each realized by a trained model here. *Wigner–Weyl*: unbroken, invariant vacuum, a single fixed point and no orbit of minimisers, $\mu^2 > 0$ degenerate within each irrep by Schur — the contrastive-divergence–eroded vacuum of §4.3. *Nambu–Goldstone*: spontaneously broken, vacuum on a G-orbit, $\mu^2 = 0$ exactly along $\dim(G/\mathcal{H})$ directions — the designed unit, and the exactly-flat-direction geometry. *Pseudo-Goldstone*: explicitly broken, tilted orbit, $\mu^2 = \delta\Sigma/F^2$ small — a slow manifold rather than a flat one; analytic tilts and the self-broken MLP. Only the *classical, tree-level* Goldstone theorem is used, a statement about a Hessian. No thermodynamic limit, no phase transition, no $\hbar$, and no loops, chiral logarithms, running low-energy constants or anomalies anywhere in this paper. *Axis 2* is the map: (ε, γ, T) fixes what a given μ^2 buys, through the three bands above. What the unit stores is the *coset coordinate*, the position along the flat direction.

Fine print, next to the claims it qualifies. *(a)* The latch's infinite half-life is a $T = 0$ statement about the deterministic damped map; at $T > 0$ the coset coordinate diffuses with $D_\theta = \varepsilon T(2-\gamma)/(2F^2\gamma)$, $F^2 = M_{\mathrm{ch}} r^{*2}$ — a *diffusive register* of finite computable lifetime, verified on the trained designed checkpoints, where raising friction *lengthens* coset memory by $3.77 \pm 0.23\times$ for $\gamma : 0.05 \to 0.2$ on $5/5$ seeds, and holding only under fluctuation–dissipation-consistent noise and a Newtonian kinetic mode, never under the reference implementation's legacy noise default. All §4 results are $T = 0$. *(b)* By Sylvester's law of inertia a flat direction keeps $\mu^2 = 0$ under any inertial anisotropy: symmetry protects the write *current*, flatness protects the *register*, and map equivariance is sufficient but not necessary for a neutral memory direction — our kinetically-broken battery is a *measured* non-equivariant map that latches exactly (Appendix B). *(c)* The *tilt-as-a-designed-lifetime-dial* construction is claimed for designed geometries only, the self-broken MLP being an *observed* instance of the cell and not a dial: in a separate battery of ours on a *learned, multi-atom* store, a breaking coefficient intended as the lifetime dial moved the softest eigenvalue *the wrong way*, and a coercivity term — not the breaking — set the ceiling (Appendix B).

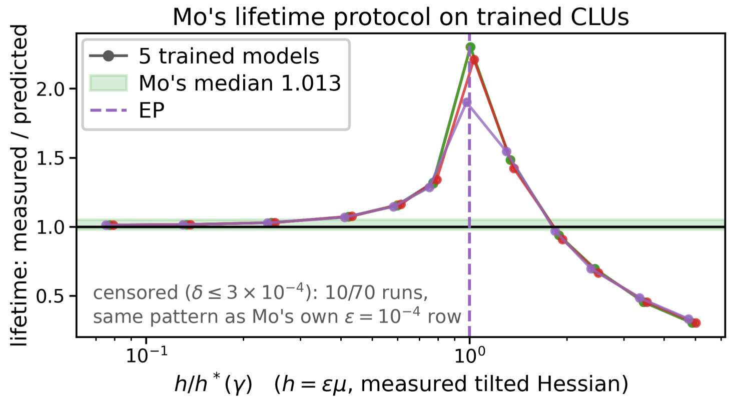

Figure 1: Mo's single-exponential lifetime law, run *unchanged* on the trained models, is the overdamped face of the transverse-curvature budget. Measured/predicted lifetime ratio versus breaking magnitude, 14 regimes, 5 seeds: ≈ 1 overdamped, reproducing his reported median 1.013; a $2.2\times$ delay spike at the exceptional point; declining to $0.31\times$ deep underdamped — containment exact below the crossover, failing above it in the calculable ballistic direction. *Evidence.*

Mandatory flag, travels with every finite-temperature number in this paper. All such results require `langevin_noise="fdt"` — the fluctuation–dissipation-consistent noise scale $\sigma_i^\star = \sqrt{M_i T \gamma(2-\gamma)}$ — and a Newtonian kinetic mode (`newtonian_identity`/`newtonian_learned`). The reference implementation's default is `legacy`, under which T is not in energy units and none of these laws hold; and in the relativistic kinetic mode no noise scale targets Gibbs at all (Appendix B; *theory note*, Prop-9′), so the laws are undefined there. The checkpoints here are `newtonian_learned`, so every finite-temperature number here is in scope. Nothing in §4 depends on it: all main-text retention results are $T = 0$ deterministic rollouts.

4 Results

Every result below lives on a trained or learned system. The budget's exactness on the designed testbed is verification and is reported in Appendix D, which establishes that 2×2 block, to machine precision, as the trained map's own angular block. The sections below ask what it buys.

4.1 Head-to-head against a recently posted lifetime law, on trained models

A zero Lyapunov exponent certifies that a direction is neutral. *It does not say how long a value written there survives once the symmetry is imperfect. We therefore run the published estimator on our trained models and read where it holds.* Mo (2026) shows that a "pseudo-gap" predicts a finite memory lifetime through a single-exponential estimator. Running his exact code-level protocol — phase 0.35 rad, threshold 0.2 rad, his censoring, cap 15000 steps — unchanged on the 5 trained models across 14 breaking magnitudes gives measured/predicted $1.012 \pm 0.000 \to 1.029 \pm 0.001$ overdamped against his reported median 1.013, 2.202 ± 0.155 at the exceptional point, and 0.309 ± 0.012 deep underdamped. His law is the overdamped face of our budget: there the two coincide, $\mathrm{corr}(\log \mathrm{pred}, \log \mathrm{meas}) = 0.9987$ overdamped-only, and his predictor fails only past the crossover — by $\approx 3.2\times$ at the deepest trained-model breaking we test, $\delta = 4$, our sweep going no deeper underdamped — in the calculable ballistic direction, with the $2.2\times$ exceptional-point delay spike as the third signature. On the *exact analytic map* the same mechanism continues to $\approx 5\times$ further underdamped (theory note, check (k), $\gamma = 0.1$); the $\approx 3.2\times$ is the trained-model number, the $\approx 5\times$ is exact-map only, and we never quote it as evidence about a trained network. This is containment, not conflict: below h^* we recover his reported number, and above it the regime structure — floor, ringing, exceptional point — is what a first-order flow cannot exhibit and what a *constitutive* curvature adds. It also does not rest on substituting the exact Jacobian gap for his predictor: the result reproduces on Mo's own finite-horizon estimator (Appendix C). *Evidence: a recently posted ML lifetime law, run on trained checkpoints.*

4.2 Designed versus emergent protection, and the price of the physics

Symmetric training data might be expected to induce a flat direction on its own. *On this architecture class it does not.* A generic MLP potential trained on that data develops *no* near-flat direction: softest emergent $\mu^2 = 5.1/5.9/5.4\times10^{-2}$ on 3 seeds against the designed flat mode $\leq 2.4\times10^{-15}$, a gap of 13–14 orders; Mo's own attribution instrument separates the architectures by 15 orders. This is a measurement on our architecture class and training recipe, not a general statement that learning cannot produce a tuned flat direction — a local learning rule that does produce one is published (Vafidis et al. 2022). Operationally that arm stores nothing on a *continuum*: $\approx$ 1–1.6 bits on 2–3 washboard minima, a written δ relaxing completely, though the damping-optimum retention law itself still generalizes to it.

The price of the prior (evidence). Against param-matched controls — dim 4, 150 epochs, $\pm 0.05\%$ params, 3 seeds, same MLP potential — the picture is Pareto, not podium. Volume conservation buys bounded (BIBO) dynamics within the coercive-potential / compact-sublevel-set scope (1.00 versus 0.33 for a broken-volume arm), a protected near-flat direction (μ^2 0.008 versus 0.122) and vacuum survival under contrastive-divergence training ($r^* = 0.72$ versus 0); leapfrog buys the latch; and the prior costs raw fit, an unconstrained twin fitting $\approx 15\times$ better, 0.0128 against 0.190 eval MSE. That cost is *contraction-forbidden* by volume conservation — the broken-volume arm recovers $\approx 2.4\times$ of it precisely by leaking volume — and is not the causal velocity cap, inactive here. The loan is called at ≈ 700 steps (measured): the twin leads by 1–2 orders to ≈ 500 steps, crosses at ≈ 700 and diverges to 196 at 5000, while the unit holds a bounded plateau $\approx$ 0.20–0.23. Its asset is boundedness by construction, not the lowest steady MSE — broken-volume (0.14) and LSTM (0.13) plateau below it — and of the licensed repairs only global damping buys the fit back, a learned friction *field* does not (Appendix A.2, with the non-comparability caveat its numbers require): paid mechanisms are not interchangeable.

4.3 Learned baselines, and the recipe that keeps the orbit intact

Gated and oscillatory recurrent networks fit this task well. *Fitting it is not holding it. We therefore write a value into each and let it run autonomously.* On Mo's S^1 path-integration family the baselines are trained with a learning-rate sweep, best RMSE kept, and are not strawmen: LSTM and LEM reach train-horizon RMSE 0.18/0.23 rad. Yet under Mo's *autonomous* protocol the structural triad of latch, μ^{-2} law and bounded motion of the stored coordinate is qualitatively absent in them: coRNN/LEM/LSTM forget the stored analog phase in $\approx 5.6/56/69$ map-steps, the stored coordinate randomizes past 1.2 rad, and they are *perturbation-fragile*, a 0.1 hidden kick collapsing LSTM $69 \to 2$ and LEM $56 \to 5$, while the generically-trained unit holds ≈ 263 map-steps with bounded motion of that coordinate (≈ 0.35 rad, budget law to $+12$–15%) and the designed unit never moves ($5/5$ seeds). *Evidence.*

Honest gap. The load-bearing claim is the qualitative triad, architecture- and compute-independent. The raw 263-versus-69 map-step ratio does not survive compute normalization, and we retire it as a compute claim: per step the Verlet update costs $\approx 6.2\times$ an LSTM cell and $\approx 3.1\times$ a LEM cell in wall time, $\approx$ 14–15$\times$ FLOPs, not width-matched, so per unit of retention the unit spends $\approx 23.5\times/\approx 14.6\times$ more wall time than the baselines it out-holds (Appendix A.3). Lacking native velocity ingestion, it cannot enter Mo's input-driven task-RMSE axis at all, so no task-RMSE was fabricated.

The designed orbit also needs a training recipe, because the default one destroys it. This is an instance of what the attractor literature calls the fine-tuning problem, the perturbation here being the training objective itself. Wake–sleep contrastive divergence drives the order parameter r^* to zero, inverting the designed vacuum in $8/8$ runs at the default 1000 epochs: a symmetry-restoration transition in the sense of §3, the potential landing in the Wigner–Weyl cell with the predicted degenerate μ^2 pair, on a horizon set by sleep-update *frequency* racing the wake clamp, and specific to flat/degenerate vacua. *Scope clause, travelling with that finding:* our sweep varies chain length only over `sleep_steps`$\in \{50, 500\}$ at otherwise fixed sampler settings and does not resolve where either sits relative to this model's mixing time, so the finding is *frequency-decisive across the two chain lengths we run*, and is not a claim that chain length is irrelevant to contrastive-divergence learning in general.

A $V(\text{data})$ energy anchor holds the condensate up — $\lambda = 100$: $5/5$ seeds, $r^* = 0.911 \pm 0.016$, at the cost of weaker noise rejection and $\approx 35\times$ higher wake MSE. That a corrective term can keep a flat direction alive is not new (Renart, Song & Wang 2003); what we add is that every headline law of the budget still holds under the correction, at $\approx 20\times$ the erosion horizon (Figure 2): 3000 anchored epochs, 3 seeds, ratio exact to 1.5×10^{-12} over 4.6 decades, slope -0.956, same floor, exceptional-point onset bit-identical at 0.5165. The architecture *breaks* the symmetry, the objective *restores* it, the anchor *keeps it broken*.

5 Discussion

Scope box — limitations and scope choices, stated so no reader has to infer them. All results are laptop-CPU, dim 4, the S^1 testbed, ≤ 5 seeds; the solvable core is quadratic and trained statements are local at a critical point of V_θ, with anharmonic deviations measured as a zero-kick retention bias of $+5$ to $+29\%$, kick-independent for $2/3$ emergent seeds and amplitude-dependent for the softest-μ^2 seed. The head-to-head is on *autonomous retention* in map-application units, and the unit does not enter the input-driven task-RMSE axis. BIBO boundedness is evidenced within the coercive-potential / compact-sublevel-set scope and concerns the autonomous, designed setting measured here — nothing here claims that symplectic structure buys off-distribution stability in a *trained, input-driven* setting, which we do not test and where a published symplectic architecture already holds the ground with a gradient bound — and friction never stabilizes a saddle (Appendix B). The symmetry framing is tree-level only. The finite-T diffusion law holds on both the designed channel (verification) and the emergent arm (evidence, 3 seeds, above $T^\star \approx 3\times10^{-3}$) under FDT-consistent noise in a Newtonian kinetic mode; the one face that does not generalize is the continuous latch register, which is designed-only. *(i) Scale is out of scope by choice:* every number here is dim 4, hidden 64, ≤ 5 seeds, single-unit, laptop-CPU, and we report no result at larger dimension, wider layers, more units or accelerator scale. *(ii) No external benchmark is won on its own headline metric anywhere in this paper*; the comparisons are diagnostic retention protocols on a synthetic S^1 family and matched-parameter ablations, not leaderboard results — and §4.3's honest gap is part of the claim, not a caveat to it. *(iii) The substrate scope, once, in our own voice: these laws govern the store; end-to-end performance additionally depends on the encoder, measured separately, with its parameter and state-byte budget ledgered.*

A price list changes what a flat direction is for. Read this way a trained memory has latches — exact flat directions, the moduli of V_θ — and registers, nearly-flat modes of a chosen half-life, fenced as designed-geometry in Appendix B; kinetic isotropy is then the price of an equivariant write *current*, not of the register. Named as directions, not results: the μ^{-2} law, crossover and floor re-measured at latent dimension $\gtrsim 64$ and $N \geq 8$ interacting units, where mode mixing can break the single-block reduction that makes the budget exact here; the head-to-head on a wider trained-model family; the anchored recipe at longer training and larger width; the equivariant-control wrapper that would put the unit on the supervised task axis; a learned-store re-test of the explicitly-broken cell respecting that fence; the map from (μ, γ, T) to lifetime whose $T = 0$ face is §4; and the price list re-derived beyond abelian $SO(2)$, for a torus T^2 or the non-abelian $SO(3) \to SO(2)$ coset — a direction, on which this paper offers no evidence.

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A Supplementary results

A.1 The laws survive the training-time correction

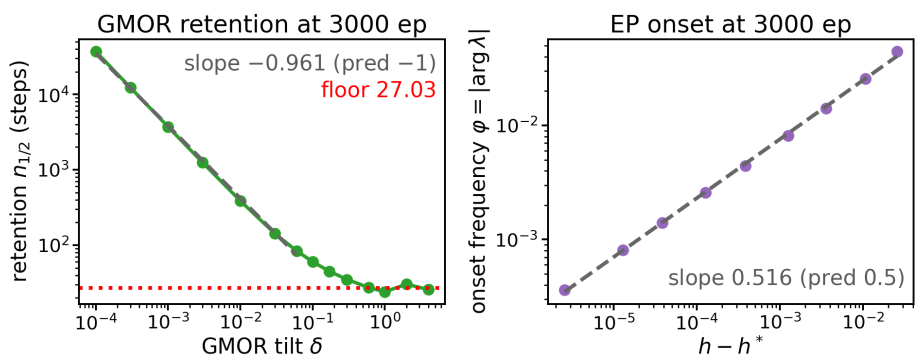

Figure 2: The headline laws of the budget survive the shipped anchor cure at 3000 epochs ($\approx 20\times$ the erosion horizon; anchored λ=100, 3 seeds). Left: the retention law $n_{1/2}$ versus tilt δ — anchored-3000-epoch points on the predicted slope, fitted -0.961 against predicted -1, saturating at the curvature-independent floor 27.03; the μ^2/δ ratio is 1.0000 ± 10^{-12} over 4.6 decades. Right: the exceptional-point onset $\varphi \propto \sqrt{h-h^*}$ above the exceptional point, fitted slope 0.516 against predicted 0.5, with $\varphi = 0$ exactly below — bit-identical to the 150-epoch value. *Designed testbed under the shipped cure; the survival of the laws through* 3000 *erosive epochs is evidence for the recipe.*

A.2 The loan curve and the recovery ladder

Dim 4*, hidden* 64*, learned-Newtonian kinetic term,* $\mathrm{dt} = 0.05$*, circle* R=1×256, 150 *epochs,* 3 *seeds, wake-only objective except the* γ_ϕ *rung. Cumulative fit MSE against horizon on the stationary circle-vacuum orbit, param-matched to* $\pm 0.2\%$ *(CLU* 4549 */ broken-volume* 4557 */ twin* 4551*).*

horizon (steps)	**twin**	**CLU**	broken-volume	LSTM
10	$\mathbf{2.9{\times}10^{-6}}$	$2.2{\times}10^{-5}$	$1.1{\times}10^{-4}$	$7.5{\times}10^{-2}$
50	$\mathbf{6.2{\times}10^{-5}}$	$1.2{\times}10^{-3}$	$1.0{\times}10^{-3}$	$4.9{\times}10^{-2}$
100	$\mathbf{2.4{\times}10^{-4}}$	$1.2{\times}10^{-2}$	$4.1{\times}10^{-3}$	$4.9{\times}10^{-2}$
500	$\mathbf{1.1{\times}10^{-2}}$	$1.8{\times}10^{-1}$	$9.7{\times}10^{-2}$	$7.7{\times}10^{-2}$
1000	$3.2{\times}10^{-1}$	$\mathbf{2.0{\times}10^{-1}}$	$1.2{\times}10^{-1}$	$9.9{\times}10^{-2}$
5000	$1.96{\times}10^{2}$	$2.2{\times}10^{-1}$	$\mathbf{1.4{\times}10^{-1}}$	$1.3{\times}10^{-1}$

The twin is best by 1–2 orders to ≈ 500 steps, crosses above CLU at ≈ 700 and diverges to 196 at 5000; CLU is the only arm flat *by construction*, on a bounded plateau ≈ 0.20–0.23. Broken-volume (0.14) and LSTM (0.13) plateau slightly *below* CLU (0.22): "physics costs raw fit" persists as a small steady-state penalty, and CLU's asset is boundedness, not the lowest plateau.

Non-comparability caveat, to be read before using these numbers. This ladder is a **separate wake-only experiment**, and its **absolute MSE scale and its twin (**0.0047**) are not cross-comparable** with the param-matched table of §4.2 (twin 0.0128, different seeds, different training objective). **Only within-table gaps are meaningful.** In particular, do not divide across tables: the "$\approx 15\times$" of §4.2 ($0.190/0.0128$) and this table's implicit "$\approx 44\times$" ($0.2066/0.0047$) are the *same qualitative statement* measured on two different scales, not two different findings, and the 92% below is a fraction of **this** table's absolute gap (0.202) only. The $\gamma = 0$ conservative rung (0.2066) is this experiment's counterpart of the $\gamma = 0$ CLU of §4.2 (0.190).

rung	eval MSE	**% gap recovered**	BIBO	latch drift	flat μ^2
twin (no physics)	0.0047	100% (def.)	1.00	—	—
CLU conservative ($\gamma = 0$)	0.2066	0% (def.)	1.00	0.778	5.2×10^{-3}
$+\gamma$ **global** ($\gamma = 0.05$)	0.0216	**92%**	**1.00**	**0.778**	$\mathbf{5.2\times10^{-3}}$
$+\gamma_\phi$ learned field ($K = 2$)	0.2548	$\mathbf{-24}$%	1.00	0.706	2.1×10^{-2}

Each licensed mechanism is added one at a time; the gap recovered is $(\mathrm{CLU}_{\gamma=0} - \mathrm{rung})/(\mathrm{CLU}_{\gamma=0} - \mathrm{twin})$, twin gap 0.202. Global γ recovers 92% of the **absolute** gap at zero structural cost, leaving BIBO, latch and μ^2 unchanged, while still trailing this table's twin by $\approx 4.6\times$ in *ratio* (0.0216 against 0.0047); both statements must travel with the claim. The learned γ_ϕ field is a targeted-forgetting tool and does not recover fit, at -24%; that rung retrained with the sleep phase on, so the "wrong tool" verdict is robust but the -24% is not a clean single-knob delta.

A.3 Per-step compute: CLU-Verlet against LSTM and LEM

Single-core CPU, f32, XLA; this backs the retirement of the "$\approx 4\times$ longer" as a compute claim in §4.3. One CLU dissipative-Verlet step (kick–drift–kick, two $\nabla_q V$ backprops through a dim-4→64→64→1 tanh MLP, learned-Newtonian kinetic term, γ=0.05, dt=0.05) is measured against one LSTM and one LEM cell, both input 1, hidden 16, output 2, the exact retention baselines of §4.3. Wall time is the median of 7 scan-amortized repetitions over 2×10^5 steps; the naïve single-call timing is dispatch-bound at a $\approx 5\,\mu$s floor and is *not* reported.

unit	params	FLOPs/step	wall ns/step (scan-amortized)
CLU Verlet (h64, dim 4, KDK)	4549	**36148**	1554
LSTM cell (h16)	1186	2512	252
LEM cell (h16)	1186	2400	500

normalization	CLU vs. LSTM	CLU vs. LEM
raw map-steps (263 vs. 69/56)	$3.8\times$ longer	$4.7\times$ longer
FLOP-normalized	$\mathbf{54.8}\times$ more	$\mathbf{70.7}\times$ more
wall-normalized	$\mathbf{23.5}\times$ more	$\mathbf{14.6}\times$ more

Per-step ratios: CLU/LSTM $= 14.4\times$ FLOPs and $6.2\times$ wall; CLU/LEM $= 15.1\times$ FLOPs and $3.1\times$ wall. The $4\times$ **inverts** under compute normalization: per unit of retention the unit spends ≈ 15–$24\times$ the wall time and ≈ 55–$71\times$ the FLOPs of the RNN it out-holds. **Confound, flagged:** the comparison is not width-matched, CLU at hidden 64 with two gradient evaluations against baselines at hidden 16; a width-matched CLU would narrow the gap, but the sign is robust because the Verlet step requires two potential backprops, and the $263/69/56$ retention numbers come from these specific configurations. The measurement is single-core CPU at batch 1; GPU or batched throughput could shift the wall ratios, though not the FLOPs. §4.3 accordingly leads with the qualitative triad, states the per-step $\approx 6\times$ (LSTM) and $\approx 3\times$ (LEM) wall factor, and does not carry the map-step ratio as a compute claim.

B Prominent negatives

Negatives are recorded, never dropped. Each row states the claim tested, the result, and the number that settles it.

claim tested	result	number
Does a learned inertial mass isotropize on symmetric data?	No; and the breaking is invisible to the Hessian and to the latch, showing only as a bounded Noether-charge oscillation.	Log-mass split ratios 0.98 / 1.18 / 5.4 from init to 150 ep; charge amplitudes $5.4/1.6/0.082\times10^{-2}$; tied controls exactly 1.0000.
Does wake–sleep contrastive divergence preserve a designed degenerate vacuum?	No — it inverts it. The cure is a V(data) energy anchor, and the headline laws survive it (§4.3).	$r^* \to 0$ in $8/8$ runs at 1000 ep; inversion at epoch 116/442/959 by sleep frequency; anchor $\lambda{=}100$ holds $5/5$ seeds at $r^*{=}0.911 \pm 0.016$.
Does an emergent (MLP) unit have a continuous coset register?	No — the continuous register is designed-only; an emergent unit stores a few bits on discrete washboard minima (evidence, 3 seeds; §4.2).	$1 - \lvert\lambda_{\mathrm{coset}}\rvert \approx 10^{-3}$ against the designed $\leq 1.1\times10^{-15}$, ~ 12 orders; written δ retained $\leq 2.1\times10^{-3}$; capacity $\approx$ 1–1.6 bits.
Do raw $n_{1/2}$ exponents discriminate designed from emergent?	No — instrument, not physics: a matched designed control on the same grid gives the same slopes, both carrying the ℓ_θ/Δ boundary layer. No $n_{1/2}$ may be quoted without its Δ and ℓ_θ/Δ.	Designed control $\mathrm{d}\log n_{1/2}/\mathrm{d}\log T = -0.53, -0.60, -1.04$ and $\mathrm{d}\log n_{1/2}/\mathrm{d}\log\gamma = +0.78, +0.63, +0.55$; ℓ_θ/Δ up to 2.0.
Is an explicit-breaking coefficient a lifetime dial on a learned store?	No, and the first half fails in sign: raising the breaking *lowers* the softest eigenvalue, and a coercivity term, not the breaking, sets the ceiling. A designed degeneracy does not survive superposition.	$\lambda_{\min}$ runs $+0.0994 \to -8.2846$ and $+0.3291 \to -1.1980$ over four decades of breaking; hard ceiling $\tau_{\max} = \Gamma/2\alpha = 4.0$; tilt-vacuum residual 0.140–0.343 against a random-direction baseline $1/\dim = 0.167$; tangential curvature predicted 0.100, measured 0.0994.
Can this unit enter the input-driven path-integration task-RMSE axis?	No — it has no native velocity ingestion and the equivariant-control wrapper is unbuilt. No task-RMSE was fabricated.	—

claim tested	result	number
Do a learned friction field and a curvature governor compose usefully?	No — the pair is Pareto-dominated by either alone, and a learned compact-support gate under-covers without accurate placement.	Locus discovery $2/6$ before adaptive-K, $2/3$ after; best learned rejection 0.861; a λ re-sweep does not approach the oracle.
Can friction stabilize a saddle, or an energy gate catch isoenergetic escape?	No to both — a genuine scope limitation of the class. Only curvature control of V_θ and the mass-anisotropic cap $v_i^{\max} = c/\sqrt{M_i}$ bound expanding or isoenergetic modes.	$\lambda_+ > 1$ for all γ, with $\min_\gamma(\lambda_+ - 1) = 1.4\times10^{-5}$.
Is the coded momentum-coupled Langevin sampler FDT-exact in every kinetic mode?	No — exact only in the Newtonian modes; in the relativistic mode *no* noise scale gives the coded chain a Gibbs invariant at all.	Required per-mode scale $\sigma_i^\star = \sqrt{M_i T\gamma(2-\gamma)}$.
Is a mean-spectrum chaos regularizer informative?	No — it is degenerate; a chaos penalty must use a max or positive-part statistic.	—
Is `sleep_temperature` active at `sleep_friction=0`?	No — a silent knob; cite it when reproducing temperature-swept sleep runs.	$\sigma \propto \sqrt{\gamma_{\mathrm{sleep}}}$, so the swept runs are bit-identical.

A reporting caution travels with the isotropization row. The closed-form charge-oscillation amplitude is an **on-vacuum-orbit** statement; **off the orbit only boundedness holds**. The amplitude-to-split ratio is moreover **not** constant, as the exact envelope requires, since it is linear only to leading order — so no proportionality constant should be quoted as a law, and we report amplitudes and boundedness, never a "drift rate". The design rule attached to mass-tying is correspondingly about the *current*, not the register: **an anisotropic channel still latches with infinite half-life**; tie the masses for an angle-independent write current, and at $\dim(G/\mathcal{H}) \geq 2$ a degenerate pseudo-Goldstone multiplet, **not** to save the register.

Two reading rules travel with the learned-store row. *(a)* The breaking coefficient of that battery is an **explicit-symmetry-breaking magnitude**; it is **not** this paper's integrator step ε, and the two must never be conflated. *(b)* No sentence in this paper may be read as claiming that a tilt magnitude is a lifetime dial on a learned store, or that a pseudo-Goldstone lifetime can be extended without bound by shrinking the breaking: on a store with a coercive confinement $\alpha\|q\|^2$ the ceiling is $\Gamma/2\alpha$, and lowering α breaks the write. This paper's own tilt results (Appendix D) are analytic tilts on **designed** architecturally-invariant potentials, where the construction is exact; the learned-store scope is exactly what this negative removes. That battery is a *different architecture* from the trained units of this paper and shares none of their training configuration, which is why it is quoted as a scope fence and never as a measurement of the units here. What survives it is the *geometry*: the identity $\lambda_{\min} = \epsilon$ holds in the single-atom geometry in which it was derived, unit-tested to 5%, and the registered damped-mode floor $\tau = 2/\Gamma$ is confirmed, 5.0 predicted against 4.23–6.58 measured time units.

The sampler row's scope, stated precisely, because the qualifier is load-bearing. The coded damping substep is an additive Gaussian momentum kick of fixed covariance, so the chain's invariant momentum marginal is necessarily Gaussian-smoothed; for a *relativistic* kinetic term the Gibbs momentum marginal is Maxwell–Jüttner, which is not of that form (*theory note*, Prop-9′ — a characteristic-function argument needing neither $V = 0$ nor linear damping). **The failure is in the *sampler*, not in the *thermodynamics***: the configurational equilibrium

$\pi_q \propto e^{-V_\theta/T}$ is relativity-insensitive, since the momentum integral factorizes out, so a relativistic unit's equilibrium distribution is perfectly well-defined and the chain simply fails to sample it; a one-line thermostat carrying a randomized, momentum-dependent inertia equal to the relativistic mass repairs it. We therefore never assert that a relativistic unit "has no equilibrium". **This touches no result in this paper:** the trained units here are `newtonian_learned` throughout, so $\sigma^\star$ is exact for every configuration we run — the qualifier is a *scope clause* on a class-level theorem, not a correction to a measurement.

C Positioning: the four retirements in full, and the head-to-head on Mo's own estimator

The four retirements of §2, elaborated so that every number behind them is on the record.

(1) Attention and state-space models have retention-guarantee analogues. HiPPO-LegS (Gu et al. 2020) has an $O(tL/\sqrt{N})$ approximation bound, a polynomial $\Theta(1/t)$ rather than exponential gradient decay, and *exact* timescale equivariance with no discretization step to tune — exact where ours is a discrete map at fixed ε with $h = \varepsilon\mu < 2$ a hard stability limit — and we have no measurement saying a learned curvature spectrum is preferable to it. Tunable permanence is likewise occupied (EDEN, Karuvally et al. 2025); UnICORNN (Rusch & Mishra 2021b) is Hamiltonian, structure-preserving and *has* the gradient-bound theorem we lack; and the modern Hopfield update *coincides with* attention for a single head with identity projections and $\beta = 1/\sqrt{D}$ (Ramsauer et al. 2021).

(2) "Retrieval is a rollout, not a weighted sum" is not our novelty. The move is Kong, Brewer & Lai's (2024), at capacity $N_c \propto K^{1.08\pm0.01}$ for their one-hot and binary index codings and 1.17 ± 0.02 for their 2D coding. Our differences are structural: neither symplectic, Hamiltonian nor mass-structured there, and the address enters through an index channel rather than as the initial condition of the state.

(3) Continuity is not an escape hatch from the NTM/DNC lineage. Both were already fully soft (Graves et al. 2014, 2016), so the precedent transfers as a warning. Of the three DNC failure modes Csordás & Schmidhuber (2019) diagnose, key/value entanglement, which they rank first, is architectural in a memory whose address q_0 is both the key and the initial condition of the value; de-allocation aliasing maps onto any position-gated forgetting; chained-read degradation is the third. None are measured here.

(4) Not a competitor to the transformer. Never claimed, and indefensible at this scale: $\approx$ 1–1.6 bits per emergent register, no win over a trivial baseline on any external benchmark, and formal limits on fixed-size latent state for copying and retrieval (Jelassi et al. 2024) that conservation does not evade — it prevents *decay*, not *capacity*.

The head-to-head reproduced on Mo's own instrument. §4.1's result does not depend on substituting the exact Jacobian gap for Mo's predictor. With Mo's own finite-horizon estimator $\hat{\lambda}(T{=}128)$ the result holds on his instrument: $\mathrm{corr} = 0.9995$ overdamped, meas/pred 0.86–1.03, and 0.30 at $\delta = 4$ against the exact-gap 0.31.

D The transverse-curvature budget verified, and GMOR proper on trained checkpoints

Designed testbed with analytic tilts and an analytic spurion, therefore verification of the theory's exactness; 5 *seeds at* 150 *epochs plus* 3 *anchored at* 3000 *epochs, dim* 4*, laptop CPU,* $f64$ *probes, probe-only with no retraining. The retention law* $n_{1/2} \propto \mu^{-2}$ *is a Gell-Mann–Oakes–Renner (GMOR) law in the spectral mass; we use the name for the identity* $\mu^2 F^2 = \delta\Sigma$ *throughout this appendix.*

C.1 The budget, verified on trained models. On the trained designed vacuum, whose flat mode has $\mu^2 \leq 2.4{\times}10^{-15}$ and is machine-flat, the constitutive identity is exact: the Hessian-derived spectral mass equals the measured one-step-Jacobian gap to $3.2{\times}10^{-10}$ over all 70 tilt rows. The GMOR law follows, $\mu^2_{\mathrm{meas}}/\mu^2_{\mathrm{pred}} = 1.000000 \pm 5{\times}10^{-12}$ at every δ and seed across 4.5 decades, with fitted overdamped retention slope -0.985 ($n = 35$; predicted -1), saturating past the crossover at the curvature-independent floor 27.03 steps at $\gamma = 0.05$. Two second-order signatures confirm a genuine exceptional, defective point. *Retention-metric bifurcation:* envelope half-life and first-crossing time agree to 0.01% overdamped and split by $3.2\times$ at $\delta = 4$, so every reported lifetime must name its metric. *Frequency onset:* $\varphi = 0$ exactly below h^* ($15/15$ rows, a real eigenvalue pair) and $\varphi \propto \sqrt{h - h^*}$ above, holding down to $h - h^* = 2.6{\times}10^{-6}$ on a fine multiplicative grid ($h - h^* \in [-4.2{\times}10^{-3}, 2.6{\times}10^{-2}]$), fitted slope 0.5165 against the predicted $1/2$ — the slight excess is far-field bending — and prefactor 0.268 against the theory note's 2×2-normalized 0.3247, a one-line units reconciliation. Where the quality factor permits ($f = 2, 4$) the trajectory frequency matches the Jacobian frequency to 0.06–0.3%; near onset the mode decays before one period completes, so the exceptional point is spectroscopically real but dynamically silent there. The latch transport law $q_\infty = q_0 + \varepsilon p_0/(M\gamma)$ holds to $\le 1\%$ over two decades of γ, the written angle then frozen to $\le 1.2\times10^{-15}$ rad.

C.2 The damping corollary — a supporting result, deliberately not a headline. The same block fixes the half-life's dependence on the damping, and that dependence is **not** monotone: sweeping γ at fixed spectral mass, $n_{1/2}(\gamma)$ traces a V-shape of depth $\approx 8\times$ whose argmin sits at, or just below, the exact critical-damping value $\gamma^* \approx 2\varepsilon\mu$, on all 5 designed seeds. Below γ^* the mode is underdamped at the floor $2\ln 2/(-\ln(1-\gamma))$, which *shortens* as γ grows; above it the mode is overdamped at $n_{1/2} \approx 2\gamma \ln 2/[(2-\gamma)(\varepsilon\mu)^2]$, which *lengthens*; the exceptional point is where the branches meet. Two consequences: **(i)** a retention measurement quoted at one γ is uninformative about the sign of $\partial n_{1/2}/\partial\gamma$ unless the band is named, the same architecture giving both signs; **(ii)** the $\mu \to 0$ latch corner is the limit $\gamma^* \to 0$, so a Goldstone register is permanently overdamped, which is the structural reason friction cannot erase it.

C.3 GMOR proper: why a second spurion. The angular tilt $V \to V + \delta(1-\cos n\theta)$ of C.1 is *radius-independent* — measured on these checkpoints, $\max|r^*_{\rm tilt} - r^*| = 2.22\times10^{-15}$ over all (checkpoint, δ) pairs — so it measures only the **product** $\mu^2F^2 = \delta n^2$, in which the condensate degenerates to the pure number n^2 (measured: $\max|\mu^2F^2/(\delta n^2) - 1| = 1.03\times10^{-11}$): the correct instrument for verifying a power law, and provably incapable of resolving the condensate. The **linear ambient spurion** $V \to V - \delta\,(u\cdot q)$ — the exact analogue of the ChPT quark-mass term, with u in the channel plane — lets the vacuum radius run with δ, so the three GMOR objects become separately measurable on a trained checkpoint: the pseudo-Goldstone spectral mass μ^2 from the autodiff Hessian at the polished vacuum; the decay constant $F^2 = M_{\rm ch} r^{*2}$, the learned channel inertia times the vacuum radius squared and **not** the orbit radius; and the condensate $\Sigma = r^*(\delta) = -\partial E_{\rm vac}/\partial\delta$. Over 8 trained checkpoints and $\delta \in [10^{-8}, 0.3]$ (10 values, 8 decades), $\max|\mu^2F^2 - \delta\Sigma| = 1.33\times10^{-15}$ over all 80 pairs.

δ	$\Sigma = r^*$ (mean)	μ^2F^2 (mean)	max rel. dev.	max **abs.** dev.
10^{-8}	0.95186937111604	$9.5186937286\times10^{-9}$	1.10×10^{-8}	1.01×10^{-16}
10^{-7}	0.95186948174437	$9.5186948223\times10^{-8}$	1.87×10^{-9}	1.85×10^{-16}
10^{-6}	0.95187058802597	$9.5187058802\times10^{-7}$	1.87×10^{-10}	1.86×10^{-16}
10^{-5}	0.95188165067351	$9.5188165068\times10^{-6}$	2.18×10^{-11}	2.16×10^{-16}
10^{-4}	0.95199226031843	$9.5199226032\times10^{-5}$	1.34×10^{-12}	1.33×10^{-16}
10^{-3}	0.95309668422658	$9.5309668423\times10^{-4}$	2.27×10^{-13}	2.09×10^{-16}
10^{-2}	0.96398369127653	$9.6398369128\times10^{-3}$	2.69×10^{-14}	2.69×10^{-16}
3×10^{-2}	0.98732513581811	$2.9619754075\times10^{-2}$	9.66×10^{-15}	2.71×10^{-16}
10^{-1}	1.06435252346294	$1.0643525235\times10^{-1}$	1.55×10^{-15}	1.53×10^{-16}
3×10^{-1}	1.60757655208590	$4.8227296563\times10^{-1}$	1.55×10^{-15}	1.33×10^{-15}

Precision fine print, next to the claim. GMOR holds to machine precision **in absolute terms at every** δ; relative exactness ($\le 2.7\times10^{-14}$) is demonstrated for $\delta \ge 10^{-2}$. The growing *relative* deviation at small δ is **not** a failure of the law but the roundoff floor of the **autodiff Hessian**: the angular curvature is $K_{\theta\theta} = \delta/r^*$, reconstructed as a difference of $O(\|K\|)$ terms, so its absolute error is $\sim \epsilon\|K\|$ and its relative error $\sim \epsilon/\delta$. This was confirmed directly: on the analytic Mexican hat a **cancellation-free** Hessian returns relative deviation 1.1–2.2×10^{-16} at *every* $\delta \in [10^{-8}, 0.3]$, while the autodiff Hessian on the same potential returns $2.28\times10^{-8} \to 4.22\times10^{-15}$ across that range, exactly $\propto 1/\delta$. **Do not quote "2.2×10^{-16} relative" for this experiment.** The same ϵ/δ floor applies to every μ^2 probe in this paper; all other sweeps here use $\delta \ge 10^{-4}$, where the floor is $\sim 10^{-12}$ and the quoted relative agreements are sound.

C.4 Σ is genuinely measured, three ways; it runs, and the angular tilt cannot see it. $\Sigma_{\rm geom} = r^*(\delta)$; $\Sigma_{\rm HF} = u\cdot q^*(\delta)$ by Hellmann–Feynman, agreeing to $\max|\Sigma_{\rm HF} - \Sigma_{\rm geom}| = 0.0$ exactly; and $\Sigma_{\rm FD} = -{\rm d}E_{\rm vac}/{\rm d}\delta$ by central finite difference at $h{=}10^{-6}$ — the envelope-theorem definition, structurally independent of the Hessian — agreeing to $\max|\Sigma_{\rm FD}/\Sigma_{\rm geom} - 1| = 9.2\times10^{-11}$. Under the linear spurion Σ runs by $+16.1\%$ to $+210.1\%$ across the δ grid (anchored-3000-epoch models $+19.8/+17.4/+16.1\%$; designed-150-epoch models $+210.1/+169.3/+35.9/+34.4/+41.1\%$), while under the angular tilt on the *same* checkpoints it does not move at all ($\le 2.22\times10^{-15}$). With u at channel angles $\{0.0, 0.7, -2.3\}$ on one designed checkpoint at $\delta{=}10^{-2}$, the μ^2 relative spread is 6.21×10^{-15}, as exact channel invariance requires.

C.5 The leading low-energy constant is saturated by the radial resonance. With $\mu^2_{\rm LO} \equiv \delta\Sigma(0)/F^2(0)$, the theory predicts a *relative* leading-order error $x \equiv \delta/(M_{\rm ch}\mu^2_{\rm rad} f)$. Measured: the ratio is 0.99999607, the mean over 8 checkpoints at $\delta \le 10^{-5}$, maximum deviation 2.20×10^{-5}; from the measured μ^2 over $\delta \in [10^{-4}, 10^{-2}]$ it lies in $[0.97917, 0.99992]$; and corrections are first order, ${\rm d(ratio)}/{\rm d}x = -1.0504$ at small x, i.e. ratio $= 1 - O(x)$ — exactly what ChPT demands of a *leading* low-energy constant. The CLU therefore realizes ChPT's **resonance saturation of low-energy constants** exactly rather than phenomenologically, the radial (σ) mode being the saturating resonance. **Breakdown is visible and expected:** at $\delta = 0.3$ the softest-radial seed, $\mu^2_{\rm rad} = 0.670$, has $x = 0.68$ — not small — and Σ has run $+210\%$; the expansion is uncontrolled

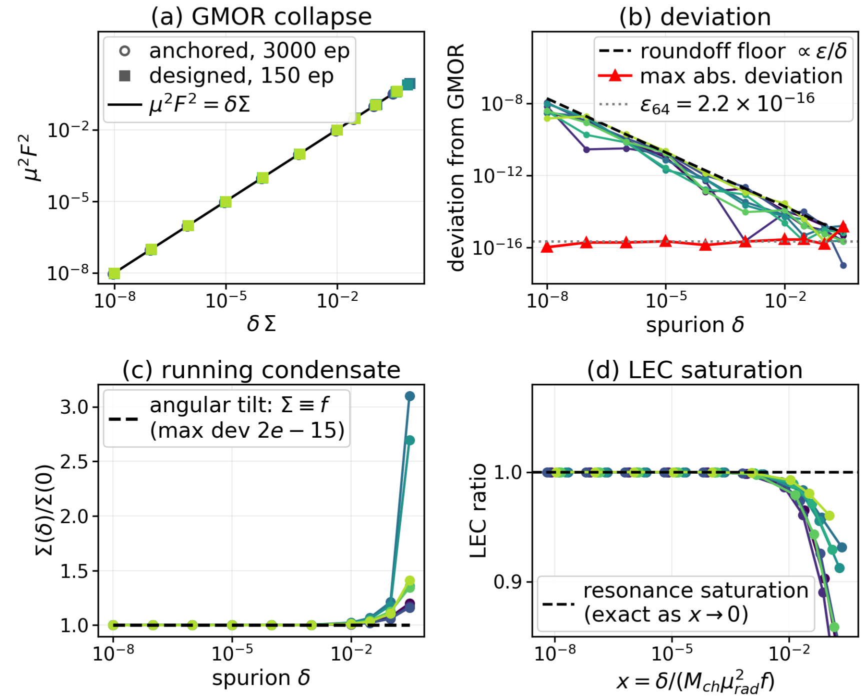

Figure 3: GMOR proper on trained checkpoints, with a linear ambient spurion. (a) The collapse $\mu^2 F^2 = \delta\Sigma$ over 8 decades of δ, 8 trained $SO(2)$ checkpoints. (b) Absolute deviation pinned at the $f64$ floor ($\leq 1.33\times 10^{-15}$) against the ϵ/δ relative-roundoff line of the autodiff Hessian — the small-δ relative residual is a probe artifact, not a law violation. (c) The running condensate $\Sigma(\delta)/\Sigma(0)$ under the linear spurion against the angular tilt's exactly flat $\Sigma \equiv r^*$ — the shipped tilt cannot see the condensate. (d) The low-energy-constant ratio tending to 1 as the chiral expansion parameter $x \to 0$, with slope -1.05. *Designed testbed, analytic spurion, trained checkpoints, therefore verification.*

there, so $\delta = 0.3$ **must not be quoted for the next-to-leading-order claim**: cap at $x < 0.25$ or state x explicitly.

C.6 Honest scope. This appendix is tree-level and classical: no loops, chiral logarithms, running low-energy constants or anomalies; "resonance saturation" is an exact algebraic statement about a two-mode classical potential, not a phenomenological fit. It is probe-only on already-trained checkpoints, spurion off by default so as to be bit-compatible with every result elsewhere, on one architecture family — designed exact $SO(2)$, dim 4, laptop-CPU. The emergent MLP arm is untested and would carry its own self-breaking δ_{eff}, so GMOR there should be tested as $\delta \to \delta + \delta_{\text{eff}}$: an open falsifiable, not a result.